

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type Mammalian, adherent

Molecules Electroporated DNA: growth hormone reporter construct with a mammalian promoter.

Species Used Human, JEG-3, choriocarcinoma cells

Before the Pulse

Cell Growth Medium MEM + 10% Fetal Bovine Serum + 1% glutamine + 1% penicillin / streptomycin (GIBCO/BRL, Sigma)

Growth Phase at Harvest 70% confluency

Pre-pulse Incubation Not given

Wash Solution MEM + 10% Fetal Bovine Serum

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 0 °C (ice)

Electroporation Medium* MEM + 10% FBS

Cuvette Gap 0.4cm

Cell Density 2.5 x 10⁸ (8) cells / ml

Voltage 0.140 kV

Volume of Cells 0.4 ml

Field Strength 0.35 kV/cm

DNA Concentration 4 µg / µl

DNA Resuspension Buffer TE Buffer (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 960 µF

Volume of DNA 12 µl

Resistor (Pulse Controller) Ω none

After the Pulse

Time Constant 50 msec

Outgrowth Medium Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature ice °C

Length of Incubation 10 min

Selection Method or Assay Used Growth hormone RIA or ELISA

Electroporation Efficiency Not given

Per Cent Survival 50 %

Name of Submitter Rich Bond, Principal Scientist

Institution Address Schering Research
Molecular Pharmacology Bldg. B-9-1
60 Orange St.
Bloomfield, NJ 07003

Survey Number

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